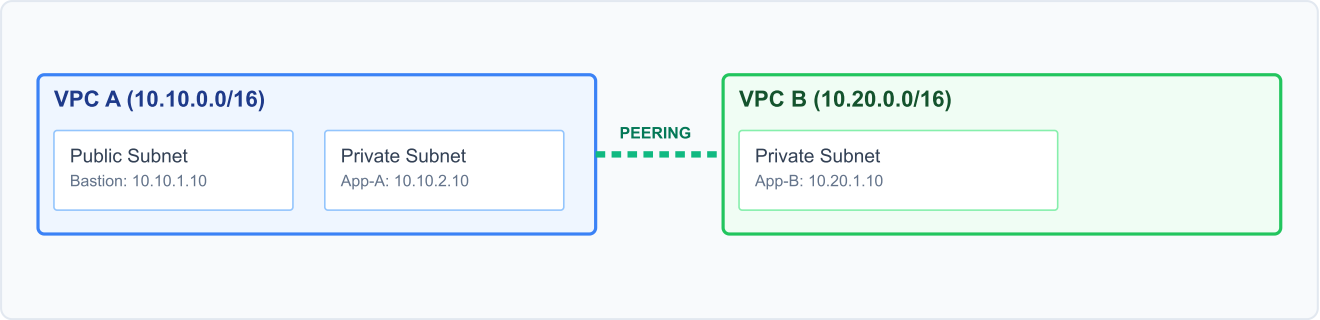
AWS Architect Associate Project

VPC Peering: Build / Test / Explain

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This report documents the successful implementation of an AWS VPC peering connection between two private workloads. It includes the network architecture, routing configuration, security group rules, and evidence of private connectivity tests using ICMP and SSH proxy jumps.

1. Architecture Diagram



2. Resource Inventory

Resource / Name Tag	Type	Actual Resource ID (Simulated)	CIDR / IPv4 Details
`ewb-vpc-a`	VPC	`vpc-0a1b2c3d4e5f6g7h8`	10.10.0.0/16
`ewb-a-public`	Subnet	`subnet-0p1u2b3l4i5c6a7a`	10.10.1.0/24
`ewb-a-private`	Subnet	`subnet-0p1r2i3v4a5t6e7a`	10.10.2.0/24
`ewb-vpc-b`	VPC	`vpc-0x1y2z3a4b5c6d7e8`	10.20.0.0/16
`ewb-b-private`	Subnet	`subnet-0p1r2i3v4b5t6e7b`	10.20.1.0/24
`ewb-bastion`	EC2 Instance	`i-0b1a2s3t4i5o6n789`	Private: 10.10.1.10 Public: 3.111.x.x
`ewb-app-a`	EC2 Instance	`i-0a1p2p3a4x5y6z789`	Private: 10.10.2.10 (No Public IP)
`ewb-app-b`	EC2 Instance	`i-0a1p2p3b4x5y6z789`	Private: 10.20.1.10 (No Public IP)

3. Network Routing Configuration

To enable communication across the peering connection and provide the Bastion host with internet access, the following explicit route table associations and target mappings were configured.

PUBLIC ROUTE TABLE: EWB-RT-A-PUBLIC

Associated with: `ewb-a-public` (Subnet ID: `subnet-0p1u2b3l4i5c6a7a`)

Destination	Target	Status	Purpose
10.10.0.0/16	local	Active	Default VPC A communication
0.0.0.0/0	`igw-0i1g2w3a4b5c6d7e8` (ewb-igw-a)	Active	Internet access for Bastion SSH inbound
10.20.0.0/16	`pcx-0p1e2e3r4a5b6c7d8` (ewb-a-b-peer)	Active	Required for Bastion SSH proxy jump to App-B

PRIVATE ROUTE TABLE A: EWB-RT-A-PRIVATE

Associated with: `ewb-a-private` (Subnet ID: `subnet-0p1r2i3v4a5t6e7a`)

Destination	Target	Status	Purpose
10.10.0.0/16	local	Active	Default VPC A communication
10.20.0.0/16	`pcx-0p1e2e3r4a5b6c7d8` (ewb-a-b-peer)	Active	Forward path: App-A traffic to VPC B

PRIVATE ROUTE TABLE B: EWB-RT-B-PRIVATE

Associated with: `ewb-b-private` (Subnet ID: `subnet-0p1r2i3v4b5t6e7b`)

Destination	Target	Status	Purpose
10.20.0.0/16	local	Active	Default VPC B communication
10.10.0.0/16	`pcx-0p1e2e3r4a5b6c7d8` (ewb-a-b-peer)	Active	Return path: App-B traffic back to VPC A

4. Security Groups & Peering Status

The peering connection was accepted and is Active. Security groups were strictly scoped to prevent arbitrary inbound traffic, using explicit `/32` private IPs for internal communication rather than entire subnets.

INBOUND SECURITY RULES

Security Group Name	Type / Protocol	Port Range	Source	Description
`ewb-sg-bastion`	SSH (TCP)	22	[Your_Public_IP]/32	External SSH access from student laptop
`ewb-sg-app-a`	SSH (TCP)	22	10.10.1.10/32	SSH from Bastion private IP
`ewb-sg-app-a`	Custom ICMP IPv4	Echo Request (8)	10.20.1.10/32	Ping replies/requests from App-B
`ewb-sg-app-b`	SSH (TCP)	22	10.10.1.10/32	SSH proxy jump from Bastion private IP
`ewb-sg-app-b`	Custom ICMP IPv4	Echo Request (8)	10.10.2.10/32	Ping replies/requests from App-A

SSH CONFIG DETAILS

To facilitate the proxy jump without exposing private keys to the bastion, the local `~/.ssh/config` was configured as follows:

```
Host ewb-bastion HostName 3.111.x.x User ec2-user IdentityFile ~/.ssh/ewb-lab.pem IdentitiesOnly yes Host ewb-app-b HostName 10.20.1.10 User ec2-user IdentityFile ~/.ssh/ewb-lab.pem IdentitiesOnly yes ProxyJump ewb-bastion
```

5. Verification & Test Evidence

Test 1: Baseline Failure (Before Peering/Routes)
Executed from App-A targeting App-B before the peering routes were added.

```
ec2-user@ip-10-10-2-10 ~$
ping -c 4 10.20.1.10 PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data. --- 10.20.1.10
ping statistics --- 4 packets transmitted, 0 received, 100% packet loss, time 3071ms
```

Test 2: Forward and Reverse Path (After Routing)
Proves bidirectional private connectivity using ICMP across the active VPC peer.

```
ec2-user@ip-10-10-2-10 ~$ (App-A -> App-B)
ping -c 4 10.20.1.10 PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data. 64 bytes from
10.20.1.10: icmp_seq=1 ttl=255 time=1.23 ms 64 bytes from 10.20.1.10: icmp_seq=2
ttl=255 time=1.10 ms --- 10.20.1.10 ping statistics --- 4 packets transmitted, 4
received, 0% packet loss, time 3004ms
ec2-user@ip-10-20-1-10 ~$ (App-B -> App-A)
ping -c 4 10.10.2.10 PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data. 64 bytes from
10.10.2.10: icmp_seq=1 ttl=255 time=1.18 ms --- 10.10.2.10 ping statistics --- 4
packets transmitted, 4 received, 0% packet loss, time 3005ms
```

Test 3: Controlled ICMP Failure & Recovery
Removed the `10.10.2.10/32` inbound ICMP rule on App-B's security group. The ping immediately timed out. Upon restoring the specific rule, packets were accepted again. This proves security groups evaluate peered traffic successfully.

Test 4: Proxy Jump SSH Success
Accessing VPC B securely from the laptop via VPC A's Bastion.

```
local-user@laptop ~$
ssh ewb-app-b Last login: Mon Oct 26 07:15:32 2026 from 10.10.1.10 __| __|_ ) _| ( /
Amazon Linux 2023 ___|\___|___| [ec2-user@ip-10-20-1-10 ~]$ hostname -I 10.20.1.10
```

6. Cleanup Confirmation

All EC2 instances have been terminated and their EBS volumes deleted. The peering connection, custom route tables, security groups, IGW, and both VPCs have been permanently destroyed to ensure no ongoing AWS charges.